Marked answers

30 student answers to the three thermal energy questions, marked and explained

Ten made-up student answers for each question in the thermal energy handout, from fully correct to fully wrong. Each answer says which mistake it adds to the test set, how we marked it point by point, and why.

Keep the **mark** and the **feedback** apart: some answers earn marks under the rules but still contain a wrong idea a tutor must correct (1.5, 1.8, 2.3, 2.10, 3.5).

No examiner report comments were available for these questions, so the mistakes are not taken from real scripts. They are common misconceptions from physics teaching (heat rises, cold flows, particles expand, shiny means absorbing) plus the grey areas the lab wants to probe: keyword stuffing, second-language English, hedging and right answer for the wrong reason.

How to read each answer

`Point (B1)`	One line of the mark scheme. Each point is worth one mark and is judged on its own. Any wording that clearly makes the point earns it.
`Any three from`	Questions 1 and 2 list five possible points but give only 3 marks. Any three earn full marks; we still tick every point made, so you can see what the answer contains.
`Intent`	What kind of mistake (or trap for the marker) this answer adds to the test set.
`Grey area`	A case where two careful human examiners could give different marks. Our mark is our best reading of the rules, not a certainty.
`Mark ≠ feedback`	The answer earns marks but contains a wrong idea. The mark stays; the feedback must still correct it.

The Cambridge rules these marks rely on

- Marks are only ever given, never taken away. A right point earns its mark even if something wrong is written next to it.
- A scheme word used wrongly earns nothing. “Convection current” earns nothing if what goes round is “the heat”.
- If a student gives two statements that contradict each other, the examiner does not pick the right one: that point is lost. Other points are not affected.
- Wrong science that has nothing to do with the question is ignored.
- Spelling and grammar are not marked unless the question says so. Meaning is what counts.
- The rule about counting only the first answers applies when a question asks for a set number of things (“give two”). These questions say explain or describe, so every valid point anywhere in the answer counts, up to the maximum.

From the general marking principles of the Cambridge 0625 June 2020 mark scheme (Paper 31, pages 2 to 4). The last rule is our reading of how the principles apply to explain and describe questions.

Question 1 EXTENDED conduction

3 marks · 10 answers

During a picnic on a warm, dry day, a metal can of lemonade is wrapped in a damp cloth.

Evaporation cools the water in the cloth.

As the water in the cloth cools, so does the lemonade.

Explain how electrons transfer thermal energy through the metal of the can. [3]

Mark scheme

ANSWER	MARK
any three from:	B3
atoms / ions vibrate	
(vibrating) atoms / ions hit electrons	
electrons propelled / travelling through metal / moving through metal	
electrons hit (distant) atoms	
free electrons / delocalised electrons mentioned	

Five possible points, three marks. Any three of the five earn full marks; a fourth or fifth point adds nothing.

ANSWER 1 **Fully correct** 3 / 3

INTENT The reference answer. It makes all five points, more than the three needed. Every marker must give full marks.

> The metal has free electrons. Near the inside of the can, where it is warmer, the metal ions vibrate. The vibrating ions hit the free electrons and give them energy. The electrons travel quickly through the metal and hit ions further away, near the cold cloth, making them vibrate more.

✓ Atoms or ions vibrate (B1)	"the metal ions vibrate".
✓ Vibrating ions hit electrons (B1)	"The vibrating ions hit the free electrons".
✓ Electrons travel through the metal (B1)	"travel quickly through the metal".
✓ Electrons hit distant atoms (B1)	"hit ions further away". The scheme says atoms; ions in a metal are the same idea.
✓ Free electrons mentioned (B1)	"free electrons".

Why 3 / 3: Five points made, three needed. The extra two add nothing to the mark but nothing is lost either. (5 of 5 points made; up to 3 count, so 3 / 3.)

ANSWER 2 **Short but complete** 3 / 3

INTENT One sentence, three points, nothing else. Tests that a marker counts points, not length.

> Free electrons gain energy, travel through the metal and collide with atoms further along.

✗	Atoms or ions vibrate (B1)	Not mentioned.
✗	Vibrating ions hit electrons (B1)	Not said where the electrons get their energy from.
✓	Electrons travel through the metal (B1)	“travel through the metal”.
✓	Electrons hit distant atoms (B1)	“collide with atoms further along”.
✓	Free electrons mentioned (B1)	“Free electrons”.

Why 3 / 3: Exactly three points, so full marks. A marker that wants the ion vibrations as well is asking for more than the scheme does. (3 of 5 points made; 3 / 3.)

ANSWER 3 **Answers a different question** 1 / 3

INTENT Describes conduction by vibrations passed from neighbour to neighbour (how non-metals conduct) and never mentions electrons, although the question asks about electrons.

> The particles in the metal vibrate more when they are hot. They bump into the particles next to them and pass on the vibration, so the heat is passed along the metal from particle to particle.

✓	Atoms or ions vibrate (B1)	“particles in the metal vibrate”. We read particles as atoms here.
✗	Vibrating ions hit electrons (B1)	The particles hit other particles, not electrons.
✗	Electrons travel through the metal (B1)	No electrons.
✗	Electrons hit distant atoms (B1)	No electrons.
✗	Free electrons mentioned (B1)	No electrons.

Why 1 / 3: Only the vibration point is in the scheme. The rest is a correct description of the slower way solids conduct, but it is not what was asked. (1 of 5 points made; 1 / 3.)

GREY AREA The scheme says “atoms / ions”. A strict examiner may refuse “particles” because in a metal the electrons are particles too, which would make this 0 / 3.

ANSWER 4 **Keyword stuffing** 1 / 3

INTENT A list of the right words with no explanation of what happens. Tests whether the marker gives credit for words or for statements.

> Conduction. Free electrons, delocalised electrons, vibrating ions, collisions, energy transfer. Metals are good conductors because of free electrons.

✗ Atoms or ions vibrate (B1)	"vibrating ions" appears only as a label in a list, not as a statement about the metal.
✗ Vibrating ions hit electrons (B1)	"collisions", but not who hits whom.
✗ Electrons travel through the metal (B1)	Electrons are never said to move.
✗ Electrons hit distant atoms (B1)	Not said.
✓ Free electrons mentioned (B1)	This is the one point the scheme gives for a mention: "free electrons".

Why 1 / 3: The scheme rewards free electrons simply for being mentioned. Every other point needs the student to say what happens, and a list of words does not. (1 of 5 points made; 1 / 3.)

GREY AREA A lenient examiner might accept "vibrating ions" as the first point and give 2 / 3. Keyword stuffing is exactly where markers drift apart.

ANSWER 5 **Cold flows in** 2 / 3

INTENT The common idea that cold is a thing that moves. The electrons carry "cold" into the lemonade instead of carrying energy out of it.

> The cold from the damp cloth goes into the metal. The free electrons carry the cold through the metal into the lemonade, so the lemonade particles vibrate slower and it gets cold.

✗ Atoms or ions vibrate (B1)	Only the lemonade particles are said to vibrate, not the metal's atoms or ions.
✗ Vibrating ions hit electrons (B1)	Not said.
✓ Electrons travel through the metal (B1)	"The free electrons carry … through the metal": the electrons are said to move through the metal.
✗ Electrons hit distant atoms (B1)	Not said.
✓ Free electrons mentioned (B1)	"free electrons".

Why 2 / 3: Two scheme points are present in the words. None of the scheme lines says which way the energy goes, so the wrong direction does not cancel them. (2 of 5 points made; 2 / 3.)

GREY AREA A strict examiner may say "carry the cold" is wrong science that cancels the travelling point, giving 1 / 3. We kept 2 / 3 because the scheme line is only about the electrons moving.

MARK ≠ FEEDBACK Whatever the mark, a tutor must correct this: energy flows from the warmer lemonade out to the cooler cloth. Cold does not flow.

ANSWER 6 **Second-language English** 3 / 3

INTENT Grammar and word order of a student writing in a second language (Portuguese first). The physics is complete. Tests that the marker does not punish language.

> The metal have electrons free. The ions is vibrating and they knock the electrons, so the electrons go fast inside the metal and knock other ions far away.

✓	Atoms or ions vibrate (B1)	“The ions is vibrating”.
✓	Vibrating ions hit electrons (B1)	“they knock the electrons”: knock means hit.
✓	Electrons travel through the metal (B1)	“go fast inside the metal”.
✓	Electrons hit distant atoms (B1)	“knock other ions far away”.
✓	Free electrons mentioned (B1)	“electrons free” is free electrons.

Why 3 / 3: All five points are there in plain words. Grammar is not marked in a physics paper. (5 of 5 points made; up to 3 count, so 3 / 3.)

ANSWER 7 **Hedging between two ideas** 1 / 3

INTENT Offers two mechanisms joined by “or” and lets the marker choose. Tests the rule that an examiner never picks the right one of two contradicting statements.

> The heat is carried by free electrons moving through the metal, or maybe by the metal atoms moving through the metal to the other side.

×	Atoms or ions vibrate (B1)	Atoms are said to move through the metal, not to vibrate.
×	Vibrating ions hit electrons (B1)	Not said.
×	Electrons travel through the metal (B1)	“electrons moving through the metal” is offered, but next to “or … atoms moving through the metal”. Two answers that contradict each other: the point is lost.
×	Electrons hit distant atoms (B1)	Not said.
✓	Free electrons mentioned (B1)	“free electrons” is mentioned and nothing contradicts that.

Why 1 / 3: Only the mention of free electrons survives. The travelling point would have counted on its own, but the student cancelled it by offering a wrong alternative. (1 of 5 points made; 1 / 3.)

ANSWER 8 **Right words, wrong process** 2 / 3

INTENT Uses the right particles but explains them with convection: the electrons get less dense and rise. Tests a right answer for the wrong reason.

> Heat rises in the metal like in convection. The free electrons get hot, become less dense and rise through the metal to the top of the can.

✗ Atoms or ions vibrate (B1)	Not mentioned.
✗ Vibrating ions hit electrons (B1)	Not mentioned.
✓ Electrons travel through the metal (B1)	“rise through the metal”: the electrons move through the metal.
✗ Electrons hit distant atoms (B1)	Not said.
✓ Free electrons mentioned (B1)	“free electrons”.

Why 2 / 3: Two scheme points are in the words, even though the reason given for the movement is wrong. (2 of 5 points made; 2 / 3.)

GREY AREA Is “less dense and rise” wrong science that cancels the travelling point? It is the student’s explanation of that very point, so a strict examiner could give 1 / 3. We gave the point because the scheme only asks that electrons move through the metal.

MARK ≠ FEEDBACK Full or part marks, the idea is wrong: single electrons do not have a density, and conduction does not depend on up or down. A good test for whether the tutor’s feedback reads the reasoning, not only the mark.

ANSWER 9 **Vague, everyday answer** 0 / 3

INTENT Says that metal conducts heat, which is true, but nothing about electrons. Tests that a marker does not reward a correct but empty statement.

> Metal is a good conductor of heat so the heat goes through it easily and the lemonade gets cold.

✗ Atoms or ions vibrate (B1)	Not mentioned.
✗ Vibrating ions hit electrons (B1)	Not mentioned.
✗ Electrons travel through the metal (B1)	Heat moves, not electrons.
✗ Electrons hit distant atoms (B1)	Not mentioned.
✗ Free electrons mentioned (B1)	No electrons at all.

Why 0 / 3: True, but it only restates the question. Nothing in it explains how the energy moves. (0 of 5 points made; 0 / 3.)

ANSWER 10 **Wrong particles vibrate** 0 / 3

INTENT The electrons vibrate and pass the vibration on, as if they were the atoms. Tests a confused concept that uses the word “electrons” throughout.

> The electrons in the metal vibrate and pass the vibration on to the next electron, like a chain, until it reaches the lemonade.

× Atoms or ions vibrate (B1)	Electrons vibrate, not atoms or ions.
× Vibrating ions hit electrons (B1)	Not said.
× Electrons travel through the metal (B1)	The electrons stay in place and pass on a vibration; they do not travel.
× Electrons hit distant atoms (B1)	Electrons hit electrons, not atoms.
× Free electrons mentioned (B1)	“Electrons”, but never free or delocalised. In a chain they are the opposite of free.

Why 0 / 3: The word electrons is everywhere, but every statement about them is the wrong picture. No scheme point is made. (0 of 5 points made; 0 / 3.)

Question 2 CORE convection

3 marks · 10 answers

Fig. 2 shows a fire heating water in a metal pan.

> *[Fig. 2: a metal pan containing water, heated by a fire underneath. Labels: “water”, “metal pan”.]*

Describe how thermal energy is transferred through the water by convection. [3]

Mark scheme

ANSWER	MARK
any three from:	B3
water particles (at bottom of pan) gain thermal / internal / kinetic energy	
(water) particles move apart	
density of liquid decreases OR liquid becomes less dense	
less dense liquid rises	
causing liquid to circulate (in pan)	

Five possible points, three marks. The points follow the story in order (energy, spacing, density, rising, circulation), but they are marked separately: any three earn full marks.

ANSWER 1 **Fully correct** 3 / 3

INTENT The reference answer, with all five points in order. Every marker must give full marks.

> The water at the bottom of the pan is heated by the fire. Its particles gain kinetic energy and move further apart, so this water becomes less dense. The less dense water rises and cooler, denser water sinks to take its place, so the water circulates round the pan.

✓	Particles gain energy (B1)	“particles gain kinetic energy”.
✓	Particles move apart (B1)	“move further apart”.
✓	Water becomes less dense (B1)	“becomes less dense”.
✓	Less dense water rises (B1)	“The less dense water rises”.
✓	Water circulates (B1)	“the water circulates”.

Why 3 / 3: Five points made, three needed. (5 of 5 points made; up to 3 count, so 3 / 3.)

ANSWER 2 **Heat rises** 0 / 3

INTENT The most common convection misconception: heat is a thing that rises, and “cold” comes down. Uses the term convection current.

> Heat rises from the bottom of the pan to the top. The hot goes up and the cold comes down, making a convection current.

×	Particles gain energy (B1)	Not about the water particles.
×	Particles move apart (B1)	Not mentioned.
×	Water becomes less dense (B1)	No density.
×	Less dense water rises (B1)	It is “heat” that rises, not less dense water.
×	Water circulates (B1)	“convection current”, but what goes round is “the hot” and “the cold”, not the water.

Why 0 / 3: Every statement is about heat moving, not water moving. A keyword used wrongly earns nothing. (0 of 5 points made; 0 / 3.)

GREY AREA Some examiners read “the hot goes up and the cold comes down” as hot and cold water and give the circulation point (1 / 3). We did not, because the student says heat, not water.

ANSWER 3 **Particles expand** 3 / 3

INTENT Gets the density story right but says the particles themselves get bigger. Tests full marks for an answer that still contains a misconception.

> The water at the bottom gets hot and the particles expand, so the water becomes less dense and rises. The cold water sinks and the water keeps going round.

×	Particles gain energy (B1)	“gets hot” is about the water, not the particles gaining energy.
×	Particles move apart (B1)	The particles expand; they do not move apart.
✓	Water becomes less dense (B1)	“the water becomes less dense”.
✓	Less dense water rises (B1)	“and rises”: the less dense water rises.
✓	Water circulates (B1)	“the water keeps going round”.

Why 3 / 3: Three points are made, so full marks. The wrong reason for the density change is not one of the points, so it costs nothing. (3 of 5 points made; 3 / 3.)

MARK ≠ FEEDBACK The particles do not expand; they move faster and further apart. The mark is full, but a tutor should still correct this.

ANSWER 4 **Keyword stuffing** 0 / 3

INTENT All the right words in a list. Tests whether a marker credits words or statements.

> Convection: particles, kinetic energy, density, less dense, rises, circulates, convection current.

✗	Particles gain energy (B1)	Words only; nothing is said to gain energy.
✗	Particles move apart (B1)	Not mentioned.
✗	Water becomes less dense (B1)	“less dense”, but not what becomes less dense.
✗	Less dense water rises (B1)	“rises”, but not what rises.
✗	Water circulates (B1)	“circulates”, but not what circulates.

Why 0 / 3: The scheme points are statements about the water. A list of words does not say what happens to anything. (0 of 5 points made; 0 / 3.)

GREY AREA A lenient examiner might give one mark for “less dense”. An LLM marker matching keywords could give 3. This is the case to watch.

ANSWER 5 **Second-language English** 3 / 3

INTENT Unusual phrasing (“less heavy for same volume”, “go far from each other”) for correct ideas. Tests that the marker reads meaning, not wording.

> The water in the bottom get the heat from fire, the particles move more fast and go far from each other. So the water is less heavy for same volume and it go up. The cold water come down, it make a cycle.

✓	Particles gain energy (B1)	“the particles move more fast”: they gain kinetic energy.
✓	Particles move apart (B1)	“go far from each other”: they move apart.
✓	Water becomes less dense (B1)	“less heavy for same volume” is the definition of less dense.
✓	Less dense water rises (B1)	“it go up”: the less dense water rises.
✓	Water circulates (B1)	“it make a cycle”: the water circulates.

Why 3 / 3: All five points, in the student’s own words. “Less heavy” alone would be wrong, but “for same volume” makes it exactly right. (5 of 5 points made; up to 3 count, so 3 / 3.)

ANSWER 6 **Hedging on density** 1 / 3

INTENT Writes “less dense or denser”. Tests that a contradiction loses the point it touches and only that point.

> The hot water becomes less dense or denser, and then it rises to the top. The water goes round the pan.

✗ Particles gain energy (B1)	Not about particles.
✗ Particles move apart (B1)	Not mentioned.
✗ Water becomes less dense (B1)	“less dense or denser”: two answers that contradict each other. The examiner does not choose.
✗ Less dense water rises (B1)	The scheme point is that the less dense water rises. Once the density is contradicted, “it rises” no longer says that.
✓ Water circulates (B1)	“The water goes round the pan”.

Why 1 / 3: Only the circulation point is untouched by the contradiction. (1 of 5 points made; 1 / 3.)

GREY AREA Some examiners accept “hot water rises” for the rising point, which would make this 2 / 3. We tied the rising point to density, as the scheme wording does.

ANSWER 7 **Conduction instead of convection** 0 / 3

INTENT Describes energy passed from particle to particle by vibrations, which is conduction. Tests that a marker notices the wrong process even when the answer sounds scientific.

> The fire heats the metal pan and the heat is passed from particle to particle through the water by vibrations until all the water is hot.

✗ Particles gain energy (B1)	No particles gaining energy; heat is passed between them.
✗ Particles move apart (B1)	Not mentioned.
✗ Water becomes less dense (B1)	No density.
✗ Less dense water rises (B1)	Nothing rises.
✗ Water circulates (B1)	Nothing circulates.

Why 0 / 3: This is a description of conduction. The question asks about convection, where the water itself moves. (0 of 5 points made; 0 / 3.)

ANSWER 8 **Particles get lighter** 1 / 3

INTENT Hot water rises because its particles get lighter. Tests a right conclusion for a wrong reason.

> The water at the bottom gets hot and the particles become lighter, so the hot water floats up. The cold water sinks. This keeps going round and round.

✗ Particles gain energy (B1)	The water gets hot; the particles are not said to gain energy.
✗ Particles move apart (B1)	Not mentioned.
✗ Water becomes less dense (B1)	Lighter particles is not lower density; the mass of a particle does not change.
✗ Less dense water rises (B1)	The water floats up because the particles are lighter, not because it is less dense.
✓ Water circulates (B1)	“This keeps going round and round”.

Why 1 / 3: Only the circulation point is made correctly. The reason for rising is wrong science about the very thing the scheme asks for. (1 of 5 points made; 1 / 3.)

GREY AREA “The water at the bottom gets hot” is close to the scheme’s “water particles (at bottom of pan) gain thermal energy”. A lenient examiner might give it, making 2 / 3.

ANSWER 9 **Stops early** 2 / 3

INTENT Two correct points and then stops. Tests that a marker counts two points as two marks, not three.

> The water at the bottom becomes less dense and rises.

✗ Particles gain energy (B1)	Not mentioned.
✗ Particles move apart (B1)	Not mentioned.
✓ Water becomes less dense (B1)	“becomes less dense”.
✓ Less dense water rises (B1)	“and rises”.
✗ Water circulates (B1)	Not mentioned.

Why 2 / 3: Two points, two marks. Everything said is right; there is just not enough of it. (2 of 5 points made; 2 / 3.)

ANSWER 10 **Irrelevant wrong science** 3 / 3

INTENT A correct answer followed by a wrong statement about something else (the pan radiating). Tests the rule that wrong science unrelated to the question is ignored.

> The particles at the bottom gain kinetic energy and spread out, so the water there becomes less dense and rises. Also the shiny metal pan is a good emitter so it radiates a lot of heat.

✓	Particles gain energy (B1)	“gain kinetic energy”.
✓	Particles move apart (B1)	“spread out”.
✓	Water becomes less dense (B1)	“becomes less dense”.
✓	Less dense water rises (B1)	“and rises”.
×	Water circulates (B1)	Not mentioned.

Why 3 / 3: Four points made, three needed. The sentence about the pan is wrong (shiny surfaces are poor emitters) but it is not about convection in the water, so it is ignored. (4 of 5 points made; up to 3 count, so 3 / 3.)

MARK ≠ FEEDBACK Full marks, but the tutor should point out that shiny surfaces are poor emitters. That idea will come back in the radiation question.

Question 3 CORE radiation

2 marks · 10 answers

Fig. 3 shows a cross-section of a flask. The flask is used to keep a liquid hot. The flask has two glass walls with a vacuum between them. The surfaces of the glass walls are shiny.

> *[Fig. 3: cross-section of the flask. Labels: “plastic cap”, “double-walled glass container”, “hot liquid”, “vacuum”, “shiny surfaces”.]*

Explain how the shiny surfaces reduce the transfer of thermal energy from the hot liquid. [2]

Mark scheme

ANSWER	MARK
(shiny surfaces) are good reflectors OR poor absorbers/emitters	B1
so less thermal energy lost by radiation	B1

Two separate marks. The second reads like a consequence of the first, but the scheme writes B1, not A1, so it can be earned on its own.

ANSWER 1 **Fully correct** 2 / 2

INTENT The reference answer, with both points and a mention of infrared. Every marker must give full marks.

> The shiny surfaces are poor emitters of infrared radiation and they reflect radiation back towards the liquid, so less thermal energy is lost by radiation.

✓	Good reflector or poor emitter (B1)	“poor emitters” and “reflect radiation back”: either is enough.
✓	Less energy lost by radiation (B1)	“less thermal energy is lost by radiation”.

Why 2 / 2: Both points are there.

ANSWER 2 **Only the property** 1 / 2

INTENT Names the property of shiny surfaces but never says what it does. Tests that the two marks are counted separately.

> Shiny surfaces are good reflectors.

✓	Good reflector or poor emitter (B1)	“good reflectors”.
×	Less energy lost by radiation (B1)	Never says that less energy is lost, or that radiation is involved.

Why 1 / 2: One correct point. The student has not finished the explanation the question asks for.

ANSWER 3 **Only the effect** 1 / 2

INTENT Says less heat is lost by radiation without saying why shiny surfaces do that. Tests whether the second mark can stand alone.

> They reduce the heat lost by radiation.

✗	Good reflector or poor emitter (B1)	Nothing about reflecting, absorbing or emitting.
✓	Less energy lost by radiation (B1)	“reduce the heat lost by radiation”.

Why 1 / 2: The scheme writes both marks as B1, so the second one does not depend on the first.

GREY AREA The answer mostly repeats the question, adding only the word radiation. Some examiners would want the reason before giving anything (0 / 2). We followed the scheme’s independent B1 marks.

ANSWER 4 **Wrong process named** 1 / 2

INTENT Reflection is right, but the energy is then said to escape by conduction. Tests that the right property does not rescue the wrong process.

> The shiny surfaces reflect the heat back so it cannot escape by conduction.

✓	Good reflector or poor emitter (B1)	“reflect the heat back”: shiny surfaces are good reflectors.
✗	Less energy lost by radiation (B1)	Says conduction, not radiation. The vacuum, not the shiny surface, stops conduction.

Why 1 / 2: The first point is right. The second names the wrong way the energy travels, so it earns nothing.

ANSWER 5 **Shiny means good absorber** 1 / 2

INTENT Thinks shiny surfaces absorb heat and hold on to it. Tests a right conclusion for a wrong reason.

> Shiny surfaces are good absorbers, so they absorb the heat and keep it in, so less heat is lost by radiation.

✗	Good reflector or poor emitter (B1)	Shiny surfaces are poor absorbers, not good ones.
✓	Less energy lost by radiation (B1)	“less heat is lost by radiation”.

Why 1 / 2: The property is wrong, but the second mark is a separate point and the statement in it is true.

GREY AREA A strict examiner may say the wrong reason contradicts the conclusion and give 0 / 2. We followed the separate B1 marks.

MARK ≠ FEEDBACK Whatever the mark, the tutor must correct the idea: shiny surfaces are poor absorbers and poor emitters. This misconception will cause wrong answers elsewhere.

ANSWER 6 **Hedging** 1 / 2

INTENT Good emitters or bad emitters. Tests that a contradiction costs only the point it touches.

> Shiny surfaces are good emitters or bad emitters of radiation, so less heat is lost by radiation.

×	Good reflector or poor emitter (B1)	“good emitters or bad emitters”: two answers that contradict each other.
✓	Less energy lost by radiation (B1)	“less heat is lost by radiation” is not contradicted.

Why 1 / 2: The examiner does not pick the right half of a hedge. The second point stands on its own.

ANSWER 7 **Answers about the vacuum** 0 / 2

INTENT Explains what the vacuum does, not the shiny surfaces. Tests that a marker checks which question was answered.

> The shiny surfaces stop heat escaping because there are no particles in the vacuum, so there is no conduction or convection.

×	Good reflector or poor emitter (B1)	Nothing about reflecting or emitting.
×	Less energy lost by radiation (B1)	Conduction and convection, not radiation.

Why 0 / 2: True physics about the vacuum, but the question asks about the shiny surfaces.

ANSWER 8 **Second-language English** 2 / 2

INTENT Word order and verb forms of a second-language writer, with correct physics. Tests that the marker does not punish language.

> The surface shiny is reflecting the infra-red back in the liquid, so the heat don’t go out by radiation.

✓	Good reflector or poor emitter (B1)	“is reflecting the infra-red back”: a good reflector.
✓	Less energy lost by radiation (B1)	“the heat don’t go out by radiation”.

Why 2 / 2: Both points, clearly made.

ANSWER 9 **Keyword stuffing** 0 / 2

INTENT A list of the right words. Tests whether the marker credits words or statements.

> Radiation, infrared, reflection, emitter, absorber, shiny, silver.

×	Good reflector or poor emitter (B1)	“emitter” and “absorber” without good or poor say nothing about the surface.
×	Less energy lost by radiation (B1)	Nothing is said to be lost, or reduced.

Why 0 / 2: No statement is made, so no point is made. Compare with answer 2, where the three words “good reflectors” earn a mark because they say something about the surface.

ANSWER 10 **Keeps the cold out** 0 / 2

INTENT Shiny surfaces reflect the cold away. The “cold flows” misconception again, in a radiation question.

> The shiny surfaces reflect the cold from outside so it can’t get in to the liquid.

×	Good reflector or poor emitter (B1)	“reflect”, but what is reflected is cold, which is not radiation. A keyword used wrongly earns nothing.
×	Less energy lost by radiation (B1)	No energy loss, no radiation.

Why 0 / 2: The word reflect is right, but it is attached to the wrong idea.

GREY AREA A lenient examiner might give the first mark for “reflect” (1 / 2). An LLM marker matching keywords is likely to.

All 30 answers at a glance

Use this as the answer key when an LLM marks the same 30 answers. Rows marked as grey are cases where human examiners could reasonably disagree; score those separately from the clear cases. Rows marked in the last column need a correction in the feedback even though they earn marks.

ANSWER	WHAT IS BEING TESTED	OUR MARK	GREY AREA	MARK ≠ FEEDBACK
Q1.1	Fully correct	3 / 3		
Q1.2	Short but complete	3 / 3		
Q1.3	Answers a different question	1 / 3	yes	
Q1.4	Keyword stuffing	1 / 3	yes	
Q1.5	Cold flows in	2 / 3	yes	yes
Q1.6	Second-language English	3 / 3		
Q1.7	Hedging between two ideas	1 / 3		
Q1.8	Right words, wrong process	2 / 3	yes	yes
Q1.9	Vague, everyday answer	0 / 3		
Q1.10	Wrong particles vibrate	0 / 3		
Q2.1	Fully correct	3 / 3		
Q2.2	Heat rises	0 / 3	yes	
Q2.3	Particles expand	3 / 3		yes
Q2.4	Keyword stuffing	0 / 3	yes	
Q2.5	Second-language English	3 / 3		
Q2.6	Hedging on density	1 / 3	yes	
Q2.7	Conduction instead of convection	0 / 3		
Q2.8	Particles get lighter	1 / 3	yes	
Q2.9	Stops early	2 / 3		
Q2.10	Irrelevant wrong science	3 / 3		yes
Q3.1	Fully correct	2 / 2		
Q3.2	Only the property	1 / 2		
Q3.3	Only the effect	1 / 2	yes	
Q3.4	Wrong process named	1 / 2		
Q3.5	Shiny means good absorber	1 / 2	yes	yes
Q3.6	Hedging	1 / 2		
Q3.7	Answers about the vacuum	0 / 2		
Q3.8	Second-language English	2 / 2		
Q3.9	Keyword stuffing	0 / 2		
Q3.10	Keeps the cold out	0 / 2	yes	